

DEV_IMPL_GUIDE - DD_REP-01 Reporting Data Mart And Dashboard API

1. Implementation Objective

Implement REP-01 as the reporting backend for dashboards, exports, and reconciliation. REP-01 prevents reporting screens from directly querying operational modules for aggregate totals.

2. Start Here

Read these first:

1. DD_REP-01-Reporting_Data_Mart_and_Dashboard_API-v1.0.md
2. DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0.md
3. FOUNDATION_KAFKA.md
4. FOUNDATION_DB.md
5. FE-APP-05-Reporting_Portal-v1.0.md

3. Build Order

1. Create reporting schema/staging tables.
2. Define event ingestion consumers for priority source modules.
3. Build v1 aggregates: revenue, payments, active subscriptions, activations, tickets, work orders, stock summary, dunning.
4. Implement dashboard APIs with role/scoping checks.
5. Implement async export requests and download authorization.
6. Implement reconciliation views comparing REP totals to source counts.
7. Add freshness metadata to every dashboard response.

4. MVP Rules

- Dashboards and exports read REP-01 only.
- Source module APIs are used only for drill-down to a specific entity.
- Heavy exports run async.
- PII masking is enforced server-side.
- REP-01 is a read model, not the owner of operational state.

5. Tests

- Event replay/idempotency.
- Aggregate reconciliation for a selected day.
- Role-based report visibility.
- PII masking.
- Export lifecycle and authorization.